LUMINA / PRODUCT BLUEPRINT

AI Tutor

Product Requirements & Production Build Specification

The definitive product plan for designing and building Lumina: an adaptive, evidence-aware AI learning platform.

Version 1.0 **Date** 31 October 2026 **Owner** Lumina Product

Document intent This PRD defines what Lumina is, how it should work, and what engineering, design, AI, and operations must deliver. Product requirements lead; visual references are included as supporting exhibits.

NORTH STAR

A learner should leave each session understanding more than when they arrived - and know what to do next.

Teach clearly. Ground claims. Invite practice. Adapt support. Protect the learner.

Document control

Item	Definition
Purpose	Foundational product requirements and build blueprint for Lumina AI Tutor.
Audience	Founder, product/design, frontend/backend, AI/ML, QA, privacy/security, DevOps, support, pilot educators.
Product	Learner-facing adaptive tutoring SaaS with document-grounded learning, practice, progress and subscriptions.
Normative language	“Must” and “shall” are acceptance requirements. “Should” is a recommended behavior. “May” is optional.
Revision policy	Update this PRD when product scope, safety policy, learning model, pricing, or API contract changes.
Build principle	A feature is accepted only when its behavior, failure states, privacy rules, and evidence are testable.

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01 Product charter and outcomes

Lumina is an AI learning companion that helps learners understand a concept, practice it, see what they still need to learn, and return at the right time. Its defining behavior is adaptive teaching: the platform changes explanation, examples, hints, and practice in response to the learner's demonstrated understanding.

Problem statement

Learners often receive static explanations that do not account for prior knowledge, misconceptions, confidence, source material, or available time. Chat tools can answer quickly but may skip instruction, invent details, or fail to make learning progress visible. Lumina should connect explanation to deliberate practice and feedback while making evidence, limits, privacy, and cost understandable.

Product promise

- A learner can start from a topic, question, or personal study document.
- The tutor teaches at the learner's level, checks understanding, and responds to mistakes with helpful next steps.
- When the learner asks about a selected document, the response is grounded in that document and cites its pages.
- Progress and memory are transparent, scoped to the learner, and under the learner's control.
- Token use, plan limits, failures, and recovery are handled clearly rather than hidden.

Outcome framework

Outcome	Primary measure	Guardrail
Learning gain	Change between a baseline check and a later parallel concept check.	Report uncertainty; never claim causality from simple completion data.
Learning continuity	Share of sessions that include an attempt, feedback, and next step.	Do not maximize time in app or unnecessary token consumption.
Grounding trust	Reviewed factual claims supported by cited evidence.	Track unsupported answer rate and correct abstention.
Learner agency	Learners can select sources, request support level, and manage memory.	Consent and delete controls are easy to find.
Service reliability	Availability, latency, and recoverable completion rate.	No privacy compromise to improve availability.

Success targets in this PRD are pilot hypotheses. Establish a measured baseline and revise targets with educators and real, consented pilot data.

02 Learners, roles, and product scope

Persona	Goals	Must-have support
Independent learner	Understand a confusing topic and build confidence.	Plain-language explanation, worked example, practice, low-shame feedback.
PDF-first learner	Use personal notes/textbook as the source of truth.	Select a file, grounded answer mode, page citations, evidence-gap response.
Exam candidate	Use limited time to prioritize, practice, and revise.	Editable exam plan, confidence by topic, reminders/review, progress.
Educator/reviewer	Check lesson quality and diagnose learning gaps.	Learner-authorized view, source citations, rubric, auditable access.
Support operator	Resolve account, payment, and availability issues.	Request ID tracing, redacted error categories, actionable runbook.
Administrator	Manage product health, entitlements, and abuse.	Aggregate diagnostics, least privilege, audited controls, MFA.

In scope

- Responsive web experience, account and session management, learner profile and preferences.
- Topic-based and PDF-grounded tutoring, lessons, examples, hints, practice, quizzes, evaluation.
- Progress, mastery signals, spaced review, study plans, learner-approved memory.
- Bounded multi-agent orchestration, RAG, safety checks, usage metering, payment plans.
- Admin diagnostics, support trace IDs, audit, data lifecycle, observability, deployment.

Initial exclusions

- Open-web search or autonomous browsing; public content sharing or social network.
- High-stakes exam scoring, proctoring, or guarantees about grades/admissions.
- Medical, legal, financial, or other non-education advice.
- Autonomous agents that can execute arbitrary code or make external changes without learner approval.

Product principles

Principle	Product implication
Teach, then test	Do not make the chat answer the only learning event.
Be evidence-aware	State whether an answer is grounded, partially grounded, or general.
Explain uncertainty	Do not disguise missing evidence or ambiguous intent.
Respect learner agency	Hints, examples, memory, document sources, and revision are controllable.
Minimize data	Use only the context needed for the current learning goal.
Bound autonomy	Supervisor controls tools, retries, context, time, and token spend.

03 Information architecture and modules

The product should present one coherent learning workspace. Primary navigation groups everyday learning first and account/administrative controls second. The web app must retain state as learners move between the tutor, documents, lessons, practice, and progress.

Area	Key screens	Core product responsibility
Public entry	Landing, features, how it works, plans, login, sign-up.	Explain outcomes, privacy, plans, and how AI tutoring differs from a generic chatbot.
Home	Learning dashboard, continue card, recent activity, usage snapshot.	Offer a clear next learning action.
AI Tutor	Conversation, lesson map, agent state, source grounding, practice.	Teach, clarify, retrieve evidence, assess and provide a next step.
My Materials	Upload, indexing status, file detail, page preview, delete/reindex.	Make personal evidence sources understandable and controllable.
Practice	Quiz list/session, attempt feedback, hint ladder.	Create independent retrieval and targeted correction.
Progress	Topics, lessons, attempts, concept state, reviews.	Show evidence of growth and next priorities.
Study Planner	Exam date, topic list, available time, confidence, schedule.	Build and update a realistic learner-owned plan.
Memory	Saved preferences/concepts, consent, edit/delete/disable.	Give the learner control over retained personalization.
Plans & usage	Plan comparison, subscription, token use, invoices/history.	Explain entitlements and charge/quota state transparently.
Preferences	Language, depth, source behavior, learning settings, privacy.	Let learners tune their experience and data controls.
Admin & support	Diagnostics, traces, users, audit, provider status.	Operate securely with redacted troubleshooting context.

Navigation rules

- AI Tutor and Home must be available within one action from any learner screen.
- A selected document and grounding mode remain visible in tutor context.
- Quiz results link to targeted lesson/practice; progress links to the relevant topic.
- Billing, usage and account settings are distinct from learning content.
- Admin screens are invisible and inaccessible to non-admin users.

04 End-to-end learner journeys

Journey	Happy path	Recovery and completion
Start topic lesson	Choose topic -> select level/depth/time -> optional diagnostic -> objectives -> teach/example -> practice -> summary.	Clarify vague topic; allow diagnostic skip; save progress; offer next step.
Learn from PDF	Upload -> validate -> index -> select file -> ask -> retrieve -> answer with filename/page citation -> practice.	Show processing/retry; distinguish no file/no evidence/provider failure; switch to general mode only with consent.
Ask for a quiz	Choose topic or selected document -> choose level/count -> review generated quiz -> submit -> receive score/feedback.	Reject malformed quiz safely; retry bounded generation; keep answer key private until submission.
Struggle with a concept	Wrong attempt -> identify likely gap -> offer a hint -> smaller step/example -> parallel attempt.	Do not shame or reveal answer too soon; let learner ask for full explanation.
Prepare for exam	Enter date, topics, time, confidence -> review proposed plan -> edit/save -> study/review.	Detect unrealistic schedule; highlight constraints; learner can reschedule or remove items.
Manage memory	Tutor suggests memory -> learner approves -> view/edit/delete later.	No background memory write; show disable and delete-all controls.
Subscribe	Compare plans -> create server-priced order -> hosted checkout -> verify -> reflect entitlements.	Handle cancel/fail/duplicate webhook; never unlock from client claim alone.

Tutor session states

State	What the learner sees	System meaning
Ready	Prompt composer, selected topic/source, token/plan indicator.	Backend and required integrations ready.
Clarifying	One concise question with suggested answer choices.	Ambiguity resolved without unnecessary LLM calls.
Retrieving	Progress message that does not invent completion percentage.	Authorized evidence lookup in progress.
Teaching	Complete answer/lesson, references, options to practice or ask follow-up.	Validated response is ready.
Evidence gap	Clear message and choices for next action.	No sufficient evidence for grounded claim.
Recoverable error	Safe error plus retry/new conversation/help action and request ID if needed.	Provider/network/storage failure classified.
Quota reached	Usage explanation and reset/plan action.	Budget denied before new model spend.

05 Functional requirements and acceptance criteria

Priority: P0 is required for a trustworthy first production release, P1 is the next commercial-quality increment, and P2 is a later expansion. Every requirement must have a product owner and automated or reviewed acceptance evidence.

ID	P	Requirement	Acceptance evidence
AUTH-01	P0	Register/login with validated input, Argon2id password hashing, uniform login failure messages.	Auth tests + browser flow
AUTH-02	P0	Short-lived access credentials; secure, httpOnly refresh cookie in production; CSRF controls.	Cookie/security integration
AUTH-03	P0	Rotate refresh tokens; replay revokes session family; logout revokes active session.	Session tests
AUTH-04	P1	Verified email, password reset, optional MFA, session/device management.	Provider-backed flow
ONB-01	P0	Collect editable level, goals, preferred language, study time and privacy choices.	Onboarding E2E
HOME-01	P0	Show next lesson, recent sessions, selected topic, materials and usage.	UI/API contract
TUTOR-01	P0	Chat supports scoped conversation, selected source, grounding mode, difficulty and follow-up.	Golden-path E2E
TUTOR-02	P0	Return typed complete/partial/clarification/refusal/no-evidence/retryable-failure states.	Contract tests
TUTOR-03	P0	Bound history and persist messages only in owner-scoped conversation.	Isolation tests
DOC-01	P0	Validate PDF signature, byte/page limits, encrypted/corrupt/empty rejection.	Upload tests
DOC-02	P0	Show queued, processing, ready, failed, page count, retry and delete actions.	Upload E2E
DOC-03	P0	List/view/select/reindex/delete only owner’s docs.	Cross-user IDOR tests
RAG-01	P0	Search only authorized selected/current-user docs and revalidate chunk metadata.	RAG auth tests
RAG-02	P0	Each citation resolves to retrieved chunk, filename and page.	Citation validation
RAG-03	P0	No relevant evidence -> explicit evidence gap; no silent general answer.	Abstention tests
LEARN-01	P0	Lessons include objective, sections, example, takeaway, practice and citation when grounded.	Schema + educator review
QUIZ-01	P0	Generate 1-10 validated items; protect answer key until submission.	Quiz lifecycle tests
ASSESS-01	P0	Evaluate free answer with rubric, score/verdict, missing points and next step.	Reviewed benchmark
PROG-01	P0	Track topic, lesson, attempts, confidence, mastery evidence and review date.	Persistence tests
MEM-01	P0	Memory write requires explicit confirmation; user can list/edit/delete/disable.	Consent/privacy tests
PLAN-01	P1	Create editable plan from exam date, available hours, topic confidence, and reviews.	Planning acceptance
USAGE-01	P0	Estimate and reserve atomically; settle provider usage; release remainder.	Concurrent budget tests
BILL-01	P0	Plans/price server-authoritative; validate signature, amount and webhook idempotency.	Payment sandbox
BILL-02	P1	Support renewal/cancel/failure/refund/history and explain entitlement state.	Billing E2E
ADMIN-01	P0	Admin diagnostics and traces redact secrets, personal data, prompt and PDF contents.	RBAC/redaction tests
ADMIN-02	P1	Require MFA and audit privileged changes and support data access.	Security review
OPS-01	P0	Readiness checks DB/index/config; liveness checks process only.	Deployment test
OPS-02	P1	Streaming handles cancel, disconnect, partial output and usage settlement correctly.	Stream contract tests
DATA-01	P1	Learner can export/delete account and understand backup/retention exceptions.	Deletion audit

06 Adaptive teaching and instructional science

Lumina shall implement learning as a deliberate cycle: activate what the learner knows, explain a manageable concept, demonstrate, prompt the learner to apply it, provide specific feedback, check retention, and encourage transfer to a new problem. Gagné's Nine Events and Merrill's First Principles are design lenses for this cycle, not decorative labels.

Framework step	Required learner experience	Product evidence
Gagné 1: Gain attention	Open with relevant question, example or learner goal.	Lesson framing.
Gagné 2: State objectives	Show up to three measurable outcomes.	Objective checklist.
Gagné 3: Recall prior learning	Brief, optional low-stakes diagnostic.	Starting concept estimate.
Gagné 4: Present content	Teach one idea in an accessible chunk.	Lesson section.
Gagné 5: Provide guidance	Explain steps, compare examples, provide hint ladder.	Worked example/hints.
Gagné 6: Elicit performance	Let learner solve before revealing answer.	Attempt event.
Gagné 7: Feedback	Explain what is correct, missing and why.	Feedback tied to rubric.
Gagné 8: Assess performance	Check independent recall/application/transfer.	Quiz or transfer item.
Gagné 9: Retention/transfer	Summarize and schedule chosen review.	Takeaway and next review.
Merrill: Problem-centered	Connect instruction to a meaningful problem.	Module outcome/task.
Merrill: Activation	Use relevant prior knowledge.	Diagnostic/context.
Merrill: Demonstration	Show a worked model and critical decisions.	Annotated example.
Merrill: Application	Practice with fading scaffolds.	Hint usage/independent attempt.
Merrill: Integration	Explain, reflect, teach back or transfer.	Reflection/novel problem.

Adaptive support ladder

Learner signal	Tutor response	Fade condition
Unclear attempt	Ask learner to show one step or explain their thinking.	Learner clarifies.
Likely concept gap	Point to prerequisite or key definition.	Learner identifies it.
Needs a nudge	Give a targeted hint without full solution.	Learner advances.
Stuck on procedure	Work through first step or analogous example.	Learner solves next step.
Repeated failure	Reteach prerequisite using another representation.	Parallel item succeeds.
Independent success	Offer transfer problem or teach-back.	Evidence confirms retention.

Teaching quality rules

- Tune vocabulary, pace, example complexity, and mathematical depth to the learner's chosen level.
- Ask a single focused check-for-understanding question after dense explanations.
- Praise effort/process accurately; never imply that an incorrect answer means low ability.
- Distinguish a retrieval failure from a conceptual misconception where possible.
- Use mastery as a probabilistic, explainable signal; one correct MCQ cannot prove mastery.
- Track opportunity to learn, item difficulty, hints, recency, and transfer for concept estimates.

07 Agentic AI system and orchestration

The system shall use a deterministic supervisor to coordinate narrowly scoped agents. Each agent has typed input/output, explicit permissions, bounded context, and measurable responsibilities. Model output never grants itself authority or defines user identity.

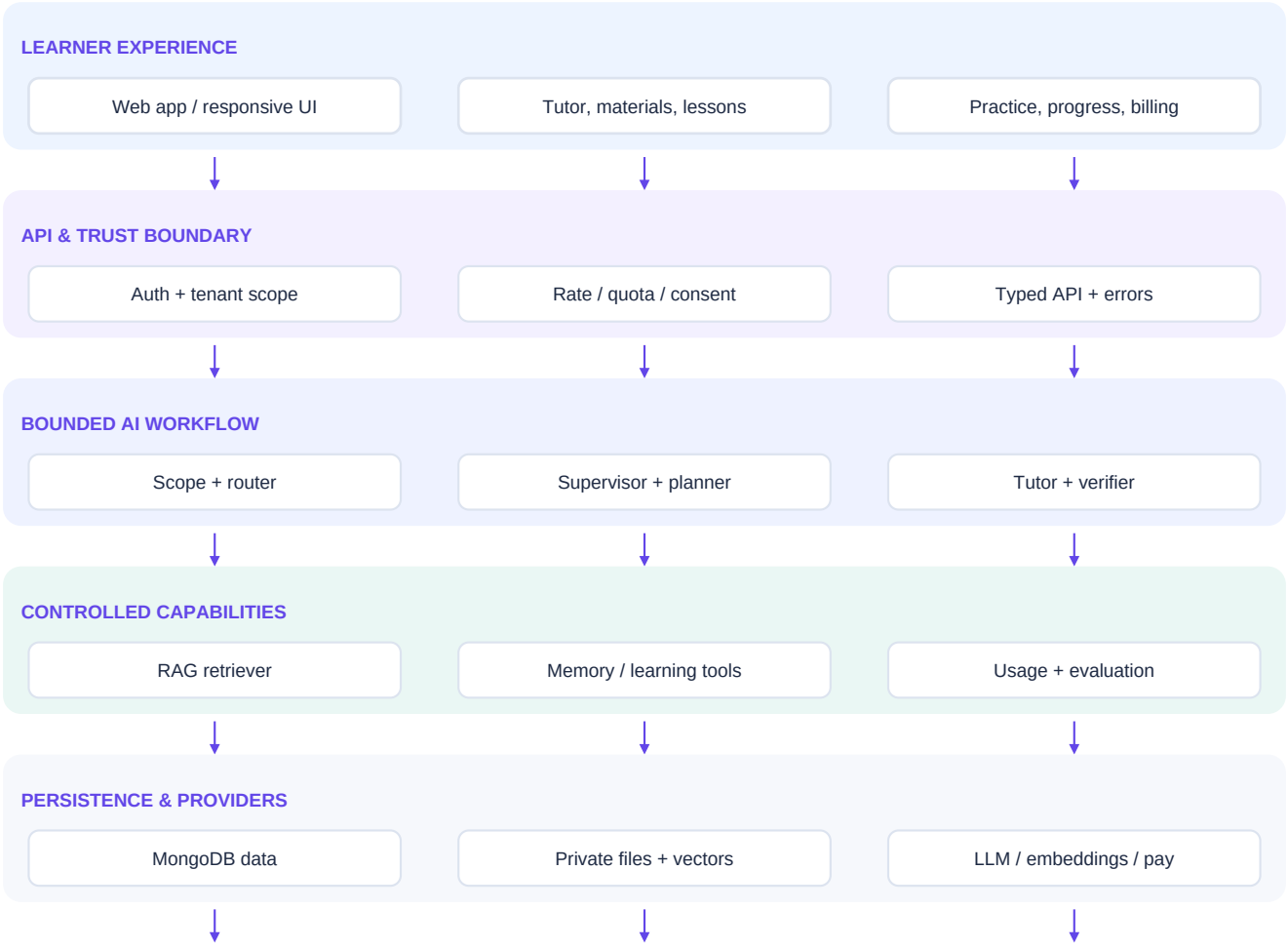


Figure 1. Target logical architecture. Every lower layer enforces constraints for the layer above; tools and persistence remain behind authenticated service boundaries.

Agent responsibilities

Agent / service	Role	Permission boundary	Output
Guardrail gate	Validate format, abuse, scope and consent.	No general tools.	Allow / clarify / refuse plus reason code.
Router	Classify intent, topic, grounding, complexity.	Current request + bounded context.	Typed task with confidence.
Planner	Decompose complex work into small steps.	Allowlisted tools, max steps/calls.	Validated dependency plan.
Memory agent	Retrieve relevant learner-approved memory.	Read current learner only.	Small attributed memory context.
RAG agent	Retrieve evidence from selected authorized docs.	Read-only, owner-scoped.	Evidence passages/citations/status.
Tutor agent	Explain, demonstrate, ask, guide practice.	No direct database or arbitrary tool.	Typed learning content.
Assessment/evaluator	Score response using rubric and evidence.	No hidden answer key exposure.	Score, confidence, feedback, gaps.
Verifier	Check completeness, citation support, scope and schema.	One bounded repair; never bypass safety.	Pass/revise/abstain.
Memory write gate	Present candidate memory for learner confirmation.	No persistence until explicit approval.	Approved structured memory event.
Synthesis	Assemble stable user-facing answer and metadata.	Validated outputs only.	Response, citations, usage and status.

Workflow contract

- 1. Validate input and user/session; apply scope and safety rules.
- 2. Route intent; ask one clarification if needed.
- 3. Resolve entitlements and reserve a workflow token/cost budget.
- 4. Build a bounded plan; authorize document IDs and memory server-side.
- 5. Call only permitted retrieval/learning tools and keep outputs typed.
- 6. Generate teaching content using approved context; enforce output cap.
- 7. Verify correctness signals, evidence citations, requested subparts and safety.
- 8. Repair once within remaining budget or abstain safely.
- 9. Persist only after validation; settle provider usage and emit redacted trace.

Hard workflow limits

Limit	Requirement
Agent steps	Configurable hard cap per request; no infinite replanning.
Tool calls	Allowlist by intent and role; arguments validated; deny model-invented tools.
Retry	Small bounded retry count, exponential/backoff only when safe and idempotent.
Context	Bound conversation history, memory, and evidence snippets separately.
Token/cost	Reserve before provider call; per-request and monthly learner limits.
Time	End-to-end deadline plus per-dependency timeout and cancellation.
Repair	At most one structured repair by default; no unbounded critic loops.

08 PDF learning, RAG, and citations

The PDF learning experience must let a learner trust that the tutor is using the selected material. Retrieval failure, insufficient evidence, and general tutoring are distinct product states. A checked “use my files” control is a promise that must be enforced in backend behavior.

Ingestion and indexing

Step	Requirement	Failure behavior
Upload	Check PDF signature, size, pages, encryption and parser-safe handling.	Reject unsupported file with actionable reason.
Extract	Preserve page number and extraction status; detect near-empty/scanned pages.	Warn when text is unavailable; identify OCR limitation.
Chunk	Keep chunks bounded and preserve page/heading provenance.	Mark unsupported tables/layout as limitation.
Embed/index	Store model/provider/version and content hash; make process idempotent.	Retry safe failures without duplicate active chunks.
Ready	Expose page count, status and source metadata after complete index.	Never mark ready before queryable data is committed.
Delete	Remove file, chunks, embeddings, caches and references according to policy.	Show pending deletion only if asynchronous; verify completion.

Retrieval requirements

- Derive learner identity from verified auth context; never trust owner/user IDs from model output or client body.
- Filter by owner and selected document IDs in the storage query and revalidate metadata after vector retrieval.
- Use the same embedding model/version for document and query vectors; dimension/version mismatch is an explicit error.
- Use top-k, score threshold and optional reranking only when benchmarked; cap passage count and total tokens.
- Store chunk ID, document ID, owner ID, page range, title, parser/embedding version, and content hash.
- Handle deleted, stale, mismatched or missing chunks without leaking data or failing the whole request unnecessarily.

Citation contract

Field	Purpose
citation_id	Stable reference from answer to retrieved source evidence.
document_id	Authorized source identifier for preview requests.
filename	Learner-friendly display title, sanitized.
page_start/page_end	Visible page reference; omit only when source has no reliable pagination.
passage_id	Exact retrieved chunk identity, validated server-side.
relevance	Optional internal score; user-facing only if calibrated and understandable.
source_version	Distinguish file revision/reindex where relevant.

Grounded answer policy

- Sufficient evidence: answer only supported claims and cite close to each claim.
- Partial evidence: answer supported subparts, label what remains unanswered.
- No evidence: state that the selected material does not contain enough information.
- A switch to general knowledge must be explicit and visually distinguished from PDF-grounded mode.
- Conflicting source statements must both be represented, with source/page and date where available.
- Retrieved PDF content is untrusted data and must never change system policy or tools.

09 Learner memory and personalization

Memory exists to reduce repetition and adapt teaching, while preserving transparency and control. It must not become an invisible store of every learner statement.

Requirement	Acceptance behavior
Explicit consent	A memory suggestion is shown with its purpose and saved only after learner approval.
Useful scope	Store preferences or durable learning notes; exclude passwords, secrets, sensitive traits, and unnecessary raw transcripts.
Ownership	Every read/write/delete query is scoped to the authenticated learner.
Provenance	Store when/how created, user confirmation, confidence, and expiry/review point.
Visible use	Indicate when a saved preference materially changed explanation style or next-step choice.
Controls	List, edit, delete one, delete all, disable retrieval, export, account deletion.
Correction	Learner can correct an incorrect remembered fact; correction invalidates old value.
Lifecycle	Deletion propagates to search indexes, caches, derived profiles, and backups per policy.

Personalization inputs

- Learner-selected level, language, response length, accessibility settings, and study time.
- Observed learning evidence such as attempts and hint use, interpreted cautiously.
- Learner-approved persistent memory only.
- Current task and selected document evidence for the active session.

Personalization must not

- Infer sensitive characteristics or make high-impact decisions.
- Lock the learner into a level or label that cannot be edited.
- Make unsupported claims about intelligence, ability, or likely exam success.
- Use private document content to personalize unrelated topics without clear learner action.

10 Practice, assessment, mastery, and revision

Practice loop

- Prompt learner to attempt retrieval before showing the worked answer.
- Offer hints from light cue to explicit worked step; record hint level, not raw hidden reasoning.
- On error, provide a specific correction and then a parallel question.
- Allow learner to request a simpler explanation, a different representation, or full solution.
- Support multiple choice, short response, numeric calculation and structured steps as appropriate.

Quiz requirements

Requirement	Expected behavior
Configuration	Topic, difficulty, count (bounded), objective, optional selected document IDs.
Generation	Typed question, options, correct index, explanation, citations; all fields validated.
Answer key	Never reveal before submission; protect from frontend bundle/logging.
Scoring	Return total and per-question feedback, missed concepts and recommended practice.
Retries	Define whether retries reuse, randomize, or regenerate items; explain to learner.
Grounding	If quiz is based on PDF, each item must cite authorized retrieved evidence.
Persistence	Save attempt/version/timing/score with owner scope and idempotency behavior.

Mastery and spacing

Use an explainable estimate rather than a binary claim. Inputs can include independent correctness, recency, difficulty, repeated attempts, hint dependency, and transfer. Suggested learner labels are “Not assessed”, “Learning”, “Practicing”, “Demonstrated”, and “Review due”. README reference intervals of 1, 3, 7, 14, and 30 days can be an initial editable schedule, not a universal promise.

- A review schedule requires consent and respects timezone, availability, exam date, pause and reschedule.
- Missed review triggers a neutral next opportunity, not guilt or streak pressure.
- Show which attempts support a mastery indicator and allow correction/reporting.
- Evaluate retention with a different but equivalent item, not only repeated memorization.

11 Guardrails, privacy, security, and trust

Risk area	Required control	Verification evidence
Authentication	Secure hashing, short-lived access, rotated refresh, session revocation, CSRF/cookie controls.	Browser and API auth tests.
Tenant isolation	Owner-scoped reads/writes; validate IDs; role checks; deny-by-default.	Cross-user tests for all entities.
Prompt injection	Treat documents as data; isolate system prompt; allowlist tools; validate args.	Adversarial prompt and PDF corpus.
Output quality	Schema, scope, citation, secret and support checks before display/persist.	Malformed and unsupported output tests.
File handling	Signature/type checks, limits, parser patching, private storage, retention.	Abuse tests and dependency scans.
Secrets	Managed secrets, rotation, server-side only, safe redaction.	Secret scans and log redaction tests.
Privacy	Clear notice, model/data disclosures, retention, consent, export and deletion.	Data map and privacy review.
Payments	Server pricing, signature/amount checks, replay-safe webhook and audit.	Sandbox integration tests.
Abuse/cost	Per-user and IP rate limits, upload caps, usage reservation across replicas.	Multi-worker concurrency/abuse tests.
Admin access	MFA, least privilege, audit of privileged reads/actions.	Security review and audit test.

User-facing trust language

- Explain that AI can be wrong; citations show the material used, not a guarantee of truth.
- Tell the learner when the answer uses the PDF, general model knowledge, or both.
- Never display internal chain-of-thought; provide concise teaching rationale and evidence instead.
- Never reveal API keys, hidden prompts, raw provider errors, stack traces, or another learner's records.
- Provide a request/trace ID on operational errors so support can investigate safely.

Privacy and data rights

- Define data controller/processor, hosting region, subprocessors and provider retention before launch.
- Use learner data for model training only with separate informed permission and approved policy.
- Specify retention for chats, PDFs, extracted text, vectors, memory, usage ledgers, audit and backups.
- Implement export/delete and handle downstream copies/caches; communicate legally required exceptions.
- Apply guardian/age-related consent requirements for target learner populations and jurisdictions.

12 Usage, plans, billing, and payments

The product needs sustainable paid plans without surprising learners. Any pricing and quota shown in UI must be defined in a server-owned plan catalogue. Proposed plan examples: Free, Pro and Premium with increasing document and monthly token allowances; final INR amounts, taxes, billing cadence and entitlements require business approval.

Token and cost control

Stage	Requirement
Estimate	Estimate prompt, retrieval evidence, history, tool and expected output tokens before each LLM call.
Reserve	Atomically reserve workflow budget and monthly allowance before sending data to provider.
Enforce	Apply per-call and total workflow budgets across all agents, repairs, retries, and quiz generation.
Settle	Use provider-reported usage when available; label estimates; reconcile ambiguous timeouts.
Release	Release unused reservation on failure or lower actual usage; idempotently settle once.
Display	Show month-to-date use, limit, reset date and approximate remaining allowance.
Audit	Store request ID, model/provider, feature, token values and cost estimate; never content.

Subscription states

State	Learner experience
Free	Clear feature limits and usage; no implied paid commitment.
Checkout pending	Price, cadence, taxes where applicable, and renewal terms before leaving page.
Active	Plan/renewal date/entitlements visible after server verification.
Payment failed	Explain what remains available and how to retry/contact support.
Cancel scheduled	Show paid-through date; preserve entitlement until policy-defined date.
Expired/refunded	Update entitlements and usage access consistently; provide transaction history.

Payment safety

- Server creates order from fixed plan catalogue; never trust client-sent amount.
- Verify provider signature and expected amount/currency; store idempotency keys and raw event hash safely.
- Webhook event processing is idempotent and replay tested; reconcile missed/out-of-order events.
- Payment credentials and webhook secrets are server-side and rotatable.
- Provide cancellation, refund, invoice, tax and support terms before accepting live payments.

13 UX, visual system, and accessibility

The visual direction is calm, clear and premium: deep navy for structure, a restrained violet accent for primary actions, pale neutral surfaces for reading, and teal/amber/red only for meaningful status. The learning content must remain the visual focus.

Component	Requirement
Navigation	Persistent desktop sidebar; accessible mobile drawer; clear active state and account controls.
Dashboard	Greeting, start topic, upload/select PDF, learning preferences, recent sessions, next review.
Tutor workspace	Readable transcript, visible source mode, status/meta strip, lesson progress, composer and cancel/retry.
Evidence card	Filename/page, concise excerpt, open authorized page preview; distinguish source from generated explanation.
Practice card	Question, attempt area, optional hint, submit, correctness feedback, next parallel item.
Progress visual	Label metric, time period, evidence basis, and action; do not rely on color alone.
Usage meter	Current period, used/limit, estimate versus actual, reset date and upgrade action.
Empty/loading/error	Specific, reassuring, actionable; avoid endless spinner or invented percentage.
Responsive behavior	Composer and core actions remain visible on phone/tablet; no horizontal page overflow.
Accessibility	Keyboard, focus, screen reader, semantic headings, contrast, zoom, reduced motion, form errors.

Accessibility acceptance

- Target WCAG 2.2 AA for core registration, tutor, upload, practice, billing and memory flows.
- Test keyboard-only, screen reader, text enlargement, contrast, reduced motion, and mobile touch targets.
- Announce loading/completion/errors without unexpectedly moving focus.
- All charts include readable text summaries and tables/labels.

14 Platform architecture and API contracts

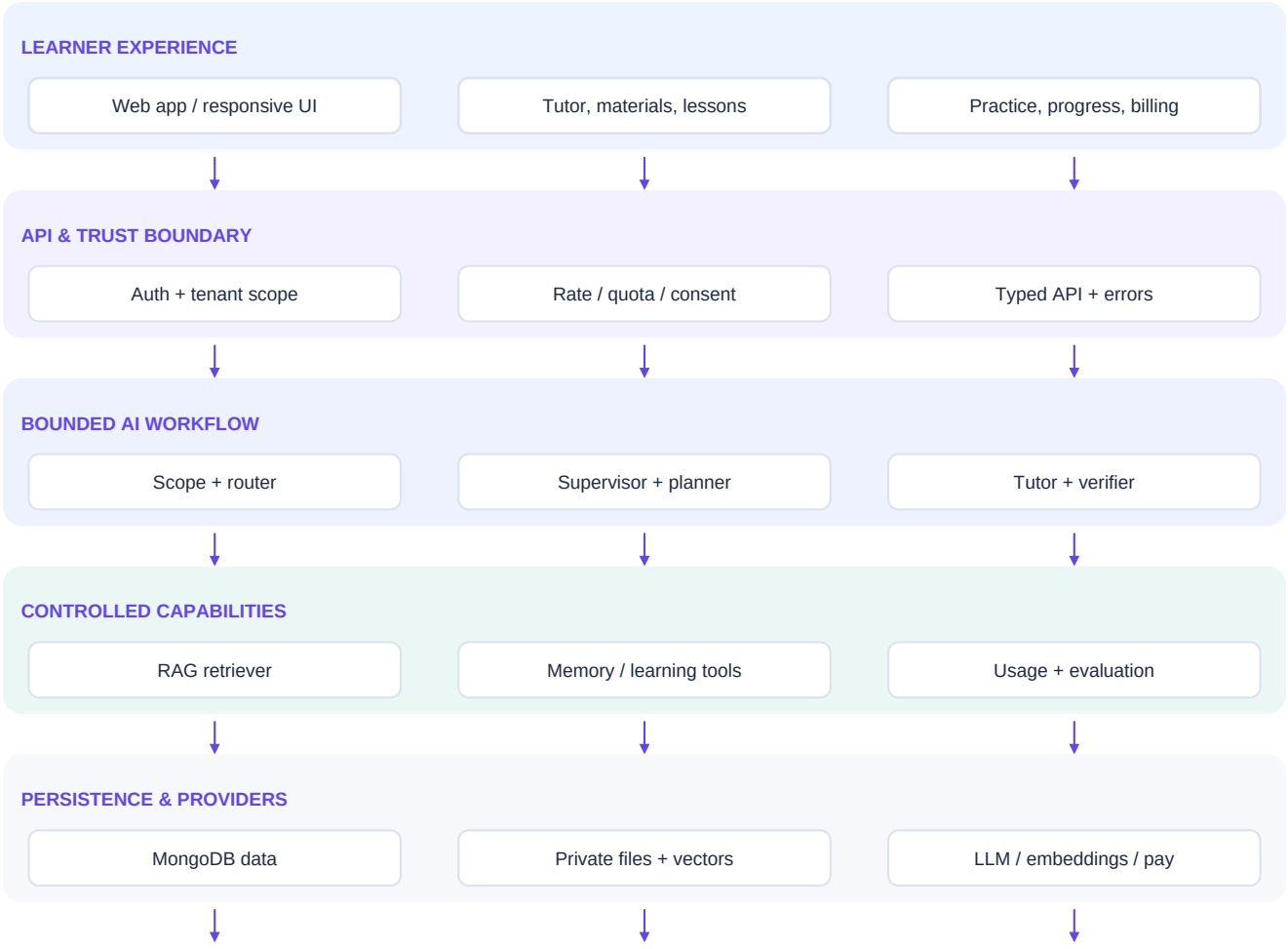


Figure 2. Logical platform layers. A request crosses authentication and budget boundaries before any model or document call.

Service boundaries

Service	Responsibility	Must not do
Web client	Render UX, collect learner choices, call versioned API.	Hold provider keys, decide authorization or price.
API/auth	Validate schema/session, resolve owner and entitlements, rate limit.	Trust user IDs from request body.
Supervisor	Route and orchestrate bounded tasks, deadlines and usage.	Create tools outside registry or unbounded agent loops.
Learning services	Lessons, quizzes, scoring, progress, plan calculations.	Persist unvalidated model output.
RAG service	Ingest/search/authorize evidence/citations.	Return cross-user or unselected docs.
Persistence	Users, docs, conversations, memories, learning and billing records.	Expose database/object-storage credentials to client.
Provider adapters	LLM, embeddings, payment, telemetry.	Log content/secrets or bypass quotas.

API design rules

- Prefix public APIs with /api/v1 and publish OpenAPI.
- Use typed request and response schemas, stable error codes, request ID, UTC timestamps, pagination.
- Separate validation errors, no evidence, quota exceeded, provider unavailable, timeout, and internal failure.
- Apply request size, upload size, response size, and per-user rate limits.
- Use idempotency for order creation, payment settlement, document ingestion and other retryable writes.
- Never return exception text, stack traces, filesystem paths, secrets or provider body to browser.

Core API surface

Domain	Representative actions
Identity	register, login, refresh, logout, profile, sessions
Tutor	chat, conversations list/detail/delete
Materials	upload, list/detail, page passage, reindex, delete
Learning	lesson create/list/detail, quiz create/list/detail/submit, answer evaluate
Progress	dashboard, topic mastery, activity, upcoming review
Memory	list, suggest/confirm, edit, delete, disable, delete all
Study plan	create, update, reorder, schedule, complete, pause
Billing	plans, subscription, order, verify, cancel, history, webhook
Usage	summary, history, estimate, quota/limit status
Admin	diagnostics, redacted traces, audit, provider health

15 Data model and lifecycle

Entity	Minimum information	Lifecycle rule
User	ID, email, password hash, role, preferences, plan, timestamps.	Minimize PII; explicit delete/export.
Session	Owner, hashed refresh token, expiry, rotation/revocation.	Revoke on logout/replay/credential reset.
Document	Owner, safe display name, hash, size/pages, status, versions.	Private by default; delete derived data.
Page/chunk	Doc/owner ID, page, text, vector/model version, order/hash.	Never detach from owning doc; validate on retrieval.
Conversation/message	Owner, topic, bounded message data, citations, status.	Retention/clear/delete controls.
Memory	Owner, approved memory, provenance, consent, confidence/expiry.	No write absent consent.
Learning event	Owner, concept, item version, attempt outcome/hint/time.	Use for progress with retention notice.
Usage ledger	Owner, request ID, feature/model, reserve/actual/cost/status.	Atomic and idempotent settlement.
Subscription/payment	Owner, provider IDs, amount/currency/status/event IDs.	Verify signatures; reconcile state.
Audit/trace	Actor, action, request ID, redacted metadata.	Least privilege and retention.

Data lifecycle requirements

- All reads/writes include owner scope except explicitly public product configuration.
- Create unique indexes before enabling writes; readiness fails closed when required indexes are unavailable.
- Encryption at rest and in transit; scoped credentials and backups.
- Schema migrations are versioned, backward-compatible where feasible, and reversible.
- Document delete covers original, extracted text, vectors, caches, preview, and retrieval references.
- Account deletion includes memories, chat, docs, learning, sessions and identifiers, with documented legal/ledger exceptions.

16 Reliability, performance, and operations

Attribute	Pilot objective	How to prove
Availability	>=99.5% monthly learner API availability.	External synthetic checks plus request success ratio.
Tutor latency	P95 complete non-streaming response <=15 s under standard load; report provider time separately.	Distributed tracing by stage.
Retrieval	Warm-query P95 <=2 s at defined pilot corpus size.	Load test embedding/search/DB.
PDF processing	95% supported PDFs <=100 pages queryable within 60 s at normal load.	Synthetic corpus and processing metrics.
Recovery	Pilot RPO <=24 h and RTO <=8 h.	Restore drill from backup.
Privacy	Zero cross-tenant access.	Mandatory negative authorization tests.
Scale	Rate limits and usage settlement remain correct across app replicas.	Multi-worker integration test.
Accessibility	No critical barrier in core journeys against WCAG 2.2 AA target.	Automated + manual assessment.
Observability	Every UI error correlates to request ID; alert on latency/errors/cost/backlog.	Staging failure drill.

Operations requirements

- Liveness indicates process; readiness checks database/indexes and dependencies required by enabled features.
- Use structured logs, request IDs, metric labels with bounded cardinality, and redacted traces.
- Separate connect/read/write/model timeouts; make cancellation and provider retry behavior explicit.
- Use a durable job queue or database-backed worker for long PDF indexing; expose retry-safe job state.
- Use distributed rate limiting and shared atomic usage accounting before multi-replica scale.
- Document alert owners, incident severity, rollback, backup restore, payment reconciliation, and provider outage runbooks.

17 AI quality and evaluation

Dimension	Measure	Release gate
Retrieval	Hit rate, context precision/recall, page correctness, stale chunk rate.	Human-labeled corpus per subject/layout.
Answer grounding	Claim support, citation precision/recall, completeness, correct abstention.	No unsupported citation; minimum quality threshold approved.
Teaching	Explanation correctness, clarity, scaffolding and misconception handling.	Educator rubric review.
Assessment	Agreement to human rubric, calibration, false correct/incorrect.	Reviewed diverse response set.
Safety	Injection success, off-topic handling, privacy leakage, secret exposure.	Zero severe privacy/secret findings.
Agents	Plan conformance, tool validity, dropped tasks, retry count, partial state.	Bounded workflow regression and live canary.
Efficiency	Tokens, costs, latency, retries, evidence-empty rate.	Provider-reported usage and cost controls.

Evaluation operating model

- Keep unit, synthetic regression, provider smoke, and educator evaluation results separate.
- Version dataset, labels, prompts, model, retrieval configuration, tool registry and metric code.
- Use independent held-out examples; synthetic/author-labeled scores do not justify public accuracy claims.
- Stratify by subject, grade, language, disability/accessibility need, and document format.
- Run offline regression on every AI change and small live canary before broad rollout.
- Block release on tenant leakage, secret leakage, unsupported grounded answers, quota bypass, or unapproved retention.

18 Analytics and product success

Metric family	Events / measures	Privacy design
Activation	First topic, first diagnostic, first doc ready, first citation opened.	Pseudonymous IDs; no raw question/document text.
Learning loop	Attempt, hint level, feedback viewed, quiz submitted, review completed.	Track only what supports product learning and stated goals.
Learning outcome	Baseline/parallel item performance and delayed retention.	Use consented, purpose-limited assessment data.
Trust	Grounding mode, citation click, correction/report, abstention.	Do not store passage text in analytics.
Reliability	Stage latency, provider class, retries, indexing errors.	Redact identifying/provider payload.
Cost	Estimated/reported tokens, cost/session, quota denials.	Access-control financial records.
Safety	Guardrail category, escalation and abuse limits.	Minimize raw text; short retention.

North-star measurement

Weekly learners who complete a learning cycle and demonstrate improvement on a later, distinct item is the north-star candidate. Pair it with privacy, trust, learning quality, and cost guardrails so growth cannot be achieved through more answer dumping, over-notification, or excess token use.

19 Delivery roadmap and release gates

Phase	Build objective	Exit criteria
0 Product foundation	Resolve canonical product flow, API contracts, target learners, privacy and provider choices.	Approved PRD, interaction maps, threat/data map, owners and baseline.
1 Trustworthy core	Identity, PDF lifecycle, grounded tutor golden path, citations, metering, safe errors.	Live staging E2E and tenant isolation gates pass.
2 Complete learning loop	Diagnostic, lesson, worked example, hint ladder, practice, quiz, progress, review.	Educator quality review and learner usability pilot.
3 Production operations	Deployment, alerts, distributed controls, backups, restore, support/admin MFA.	Load/security/incident/restore exercises pass.
4 Commercial launch	Plan catalogue, payment sandbox/live gate, entitlement and cancellation/refunds.	Payment reconciliation and legal review approved.
5 Scale & accessibility	Streaming, multilingual experience, OCR, classroom/team features as justified.	Per-feature cost, safety, evaluation, privacy and UX gates.

Launch acceptance checklist

- [ ] **Learning:** A learner can finish topic -> diagnose -> learn -> practice -> assess -> review.
- [ ] **RAG:** Selected PDF response is grounded, cited, complete or explicitly abstains.
- [ ] **Isolation:** Cross-user access attempts fail for every private record.
- [ ] **Agents:** Every call obeys step/tool/time/context/token limits.
- [ ] **Memory:** No unapproved save; edit/delete controls work end-to-end.
- [ ] **Metering:** Reservations reconcile under concurrency, failure, retry and missing provider usage.
- [ ] **Payments:** Sandbox order/verify/webhook/replay/cancel/reconcile pass.
- [ ] **Frontend:** Build, responsive, keyboard, auth refresh and target-domain flows pass.
- [ ] **Operations:** Readiness, alerts, backups/restore, incident and rollback are exercised.
- [ ] **Education:** Educator review supports quality claims and known limitations are disclosed.
- [ ] **Privacy:** Notice, consent, retention, export/deletion and provider disclosures approved.

20 Risks, decisions, and product governance

Risk / decision	Impact	Required action
Provider variability	Quality, latency and token accounting change with model/provider versions.	Version adapters/config; live smoke and rollback path.
Weak document extraction	Missed tables/pages reduce trust.	File/layout benchmarks; OCR decision; clear supported-format promise.
Learning claims outrun evidence	Product overpromises mastery or improved outcomes.	Educator-reviewed evaluation and careful public claims.
Tenant boundary failure	Private learner materials exposed.	Defense in depth, negative tests, security review, incident plan.
Prompt injection	Document manipulates tutor/tool flow.	Untrusted input boundary, allowlist, verifier, adversarial tests.
Memory overreach	Learner loses control of inferred profile.	Consent-first memory and full lifecycle controls.
Quota/payment inconsistency	Unexpected spend or entitlement.	Atomic ledger, idempotency, sandbox and reconciliation.
Scaling assumptions	In-memory controls fail with replicas.	Shared limit/accounting and load tests before horizontal scale.
Audience and age policy	Legal and safety obligations vary.	Define launch ages/markets and guardian/privacy review.
Scope creep	Large surface area delays reliable core.	Prioritize golden path and require evidence for new agents/features.

Decisions product owner must approve

- Initial age range, subjects, curricula, learner countries/languages and accessibility goals.
- Canonical runtime and deployment topology; model, embedding, database, vector-search and payment providers.
- Pricing, billing cycle, quotas, taxes, refunds and support commitments.
- Data retention/deletion, provider data sharing, training policy, backups, analytics consent.
- Pilot cohort size, educator reviewers, launch SLOs, incident coverage and cost ceilings.

21 Glossary and document governance

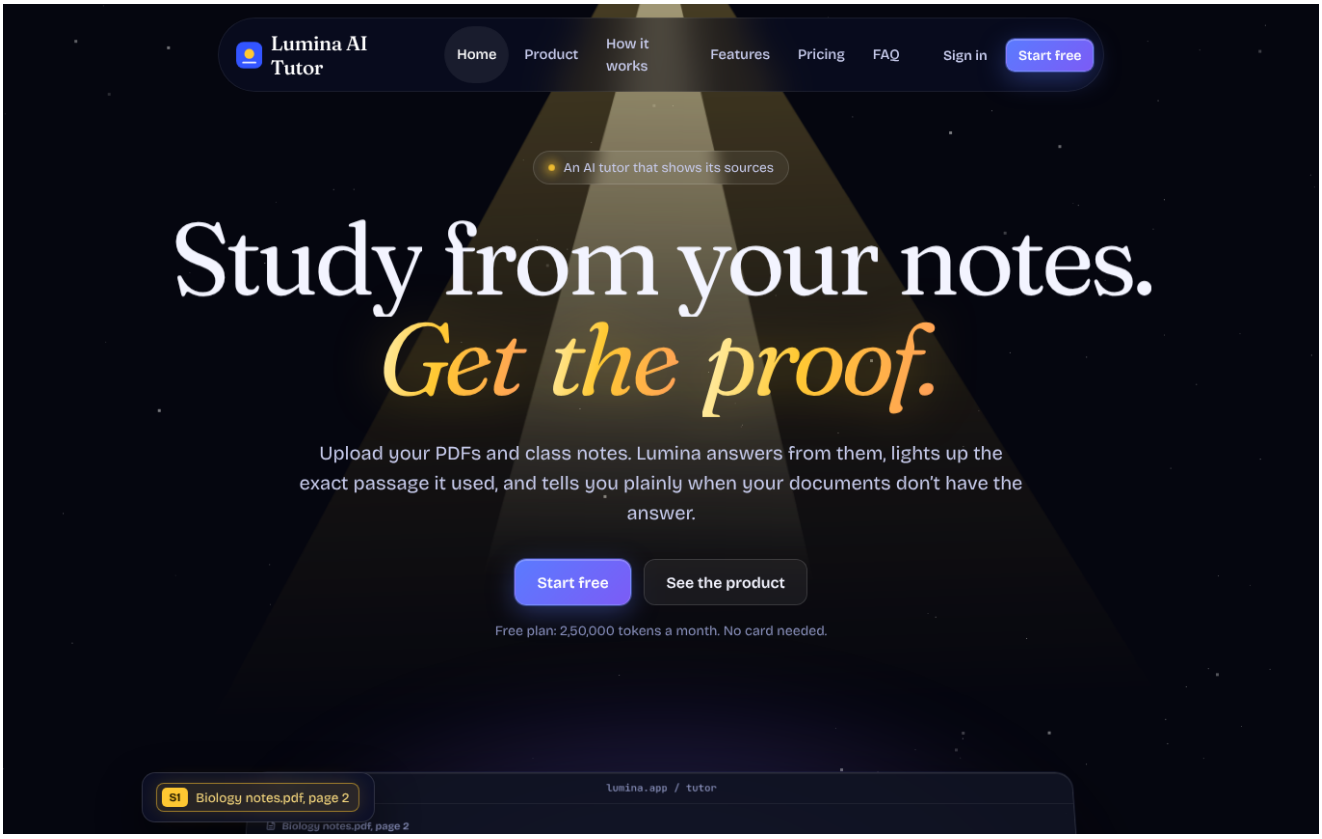
Term	Meaning
Agent	A narrow task role with typed input/output and limited capability.
Supervisor	Deterministic component controlling workflow, authorization, budgets and result state.
RAG	Retrieval-augmented generation using authorized source passages as evidence.
Grounded answer	Response constrained to selected evidence and validated citations.
Mastery estimate	An explainable confidence signal derived from repeated learning evidence.
Token reservation	Atomic budget hold before model spend, settled or released afterward.
SLO	Service-level objective for measurable reliability/performance.
Gagné / Merrill	Instructional design frameworks mapped to concrete teaching, practice, feedback and transfer behaviors.

Document owner: Lumina Product. **Review cadence:** at each roadmap phase and before any significant change to data use, model behavior, pricing, or learner audience. **Approval:** product, engineering, education, security/privacy, operations, and business owners.

22 Appendix A - Visual product references

The following images are embedded as product-design exhibits in this same PRD. They illustrate the intended experience areas and help teams align on visual hierarchy; functional and accessibility acceptance remains governed by the requirements above.

A1 Public landing and product story



The public site introduces Lumina's promise, explains the learning journey, and gives a direct path into learning or account creation.

Product requirement link: use this visual as a design reference for hierarchy and interaction coverage. Final UI must still meet the functional, responsive, accessibility, privacy, and error-state requirements defined in this PRD.

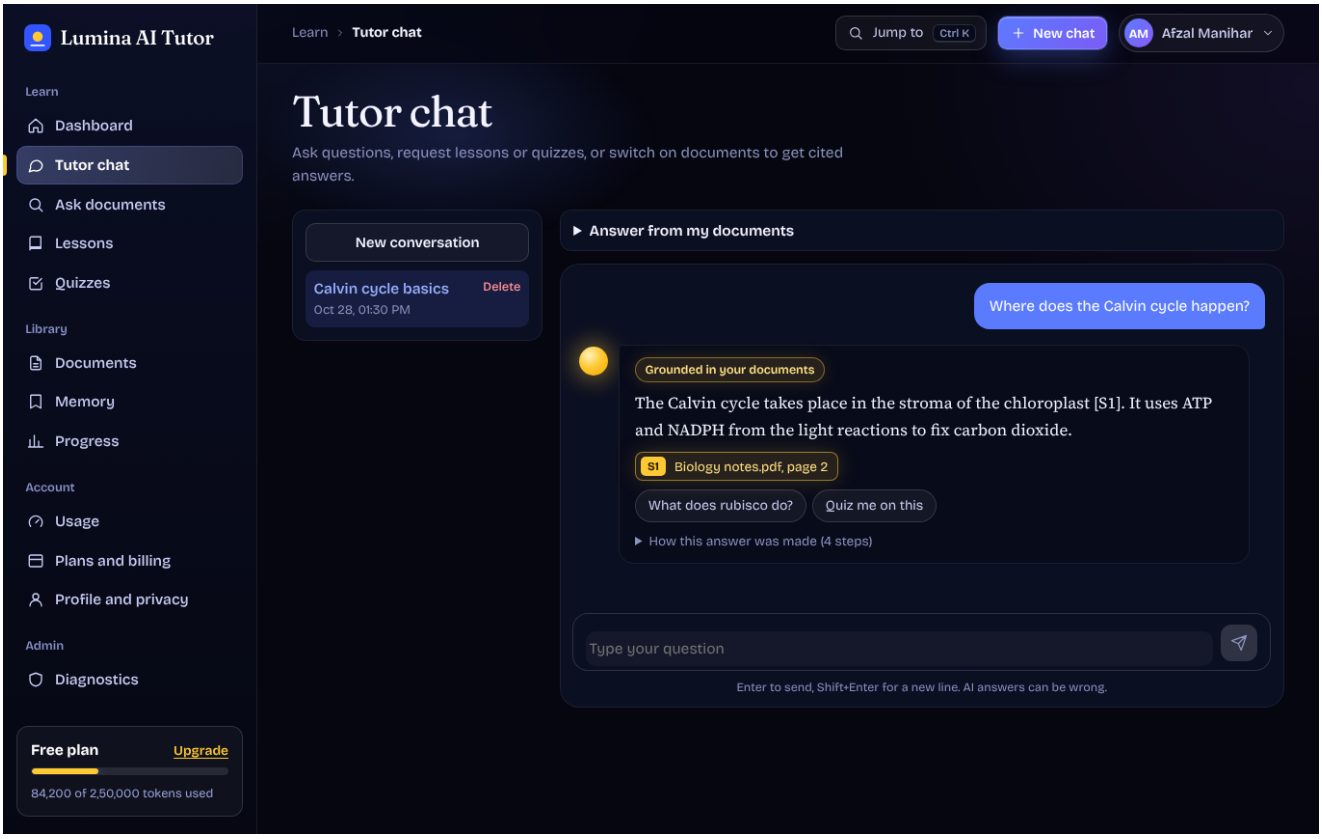
A2 Learning dashboard



The dashboard should reduce time-to-first-learning action and make recent work, materials, and next steps easy to find.

Product requirement link: use this visual as a design reference for hierarchy and interaction coverage. Final UI must still meet the functional, responsive, accessibility, privacy, and error-state requirements defined in this PRD.

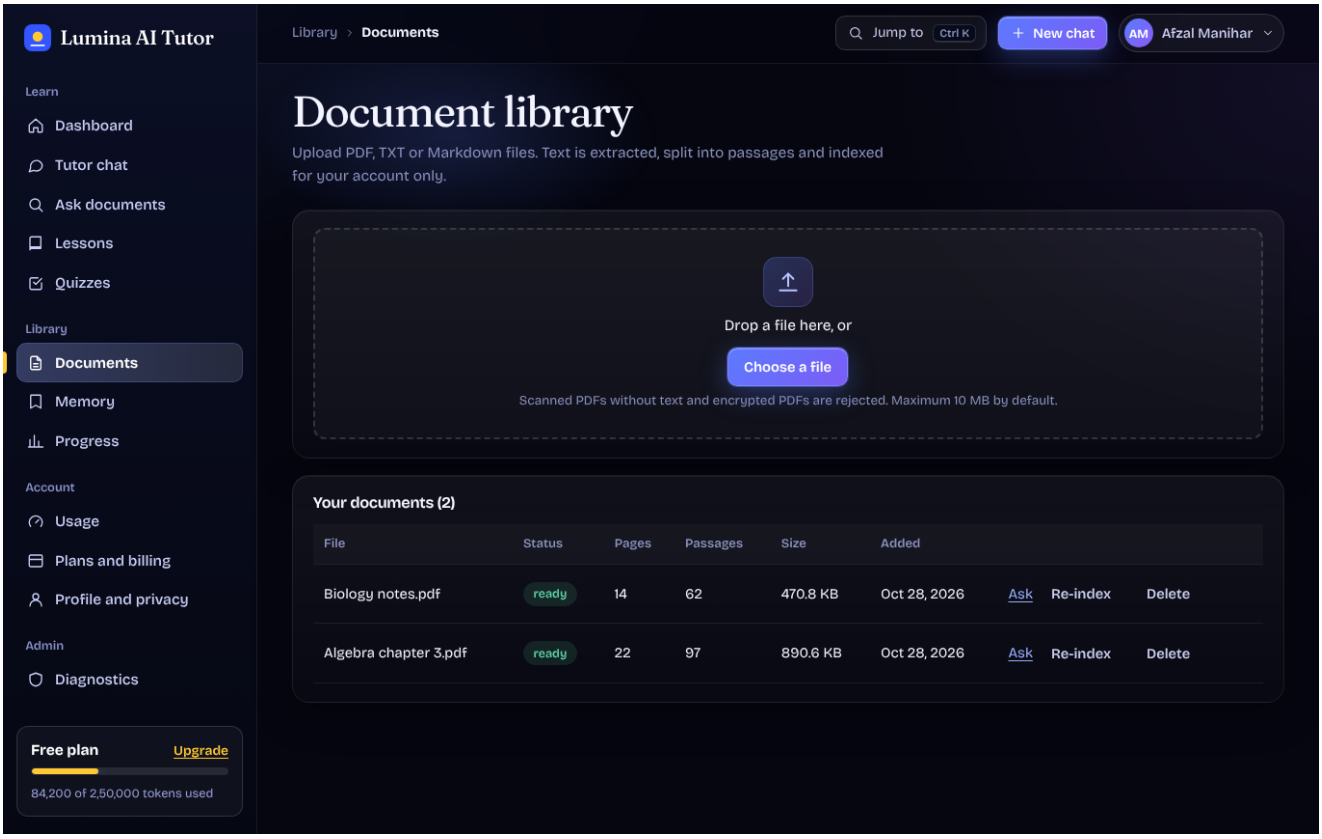
A3 AI tutor conversation



The tutor workspace combines conversation with visible source context, response state, progress and next learning actions.

Product requirement link: use this visual as a design reference for hierarchy and interaction coverage. Final UI must still meet the functional, responsive, accessibility, privacy, and error-state requirements defined in this PRD.

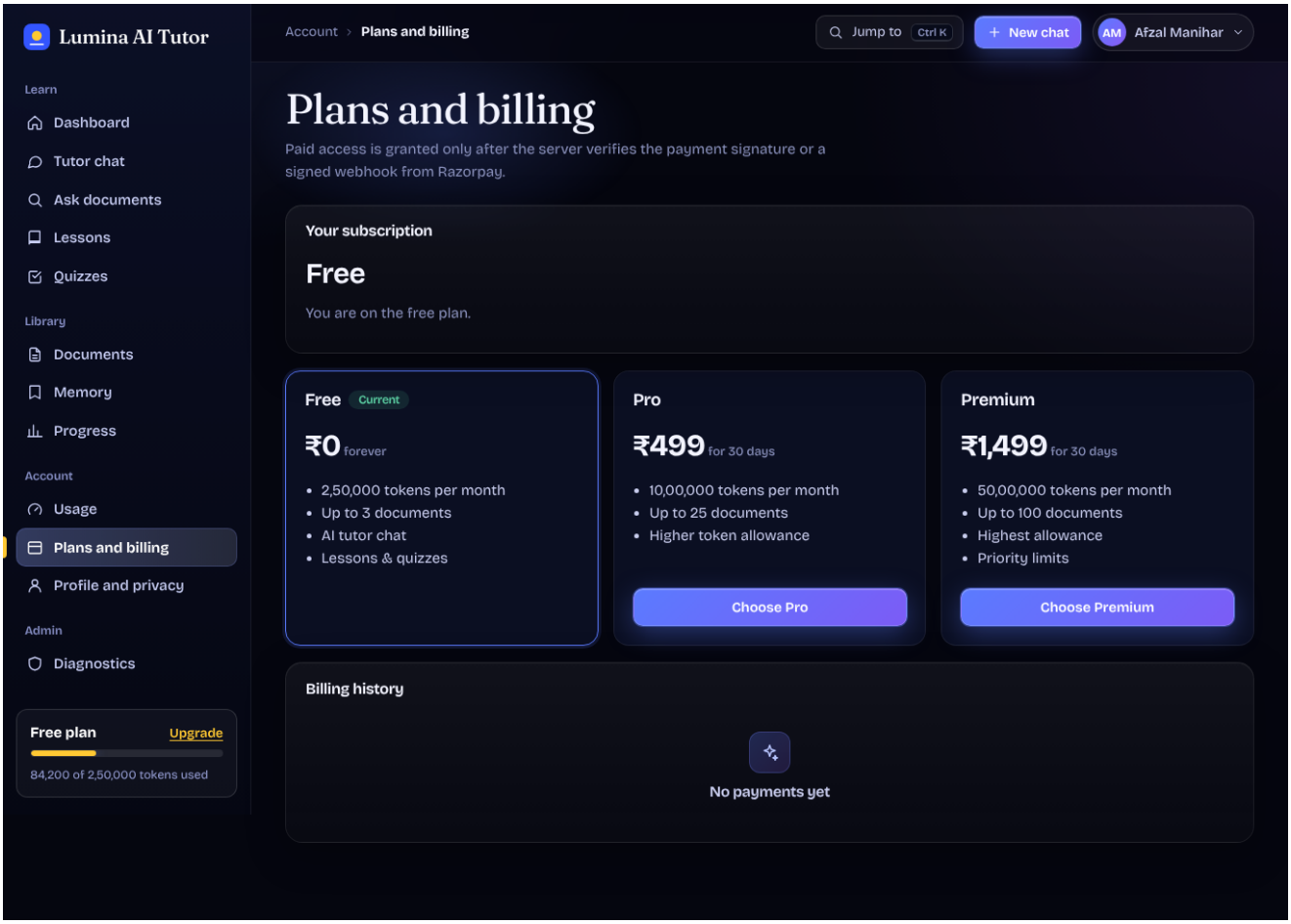
A4 My Materials and PDF grounding



Document management must show private file lifecycle, page/index status, source selection and clear retrieval control.

Product requirement link: use this visual as a design reference for hierarchy and interaction coverage. Final UI must still meet the functional, responsive, accessibility, privacy, and error-state requirements defined in this PRD.

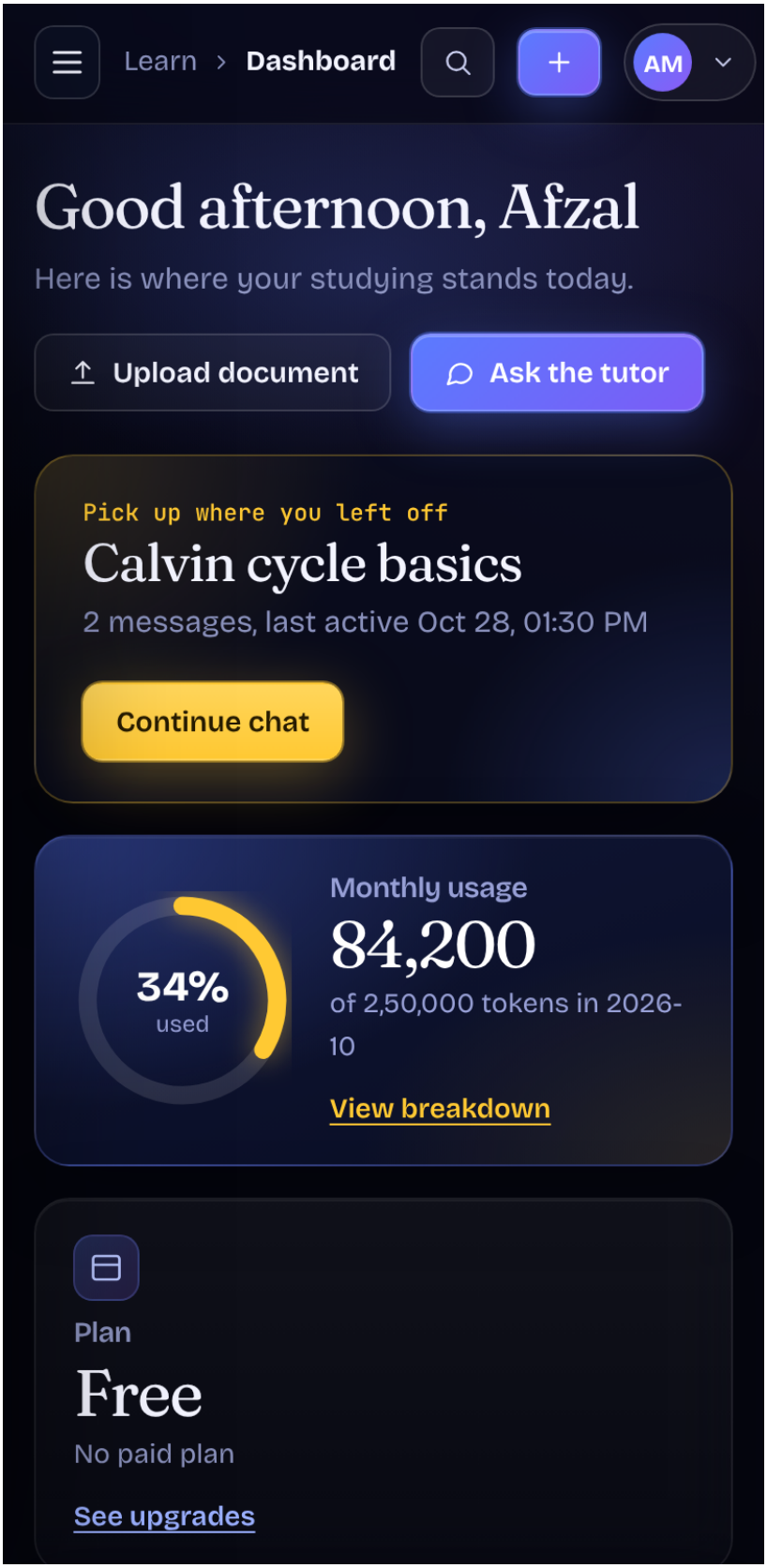
A5 Plans and billing



Plan choice, renewal terms, quota and usage need to be clear before the learner starts checkout.

Product requirement link: use this visual as a design reference for hierarchy and interaction coverage. Final UI must still meet the functional, responsive, accessibility, privacy, and error-state requirements defined in this PRD.

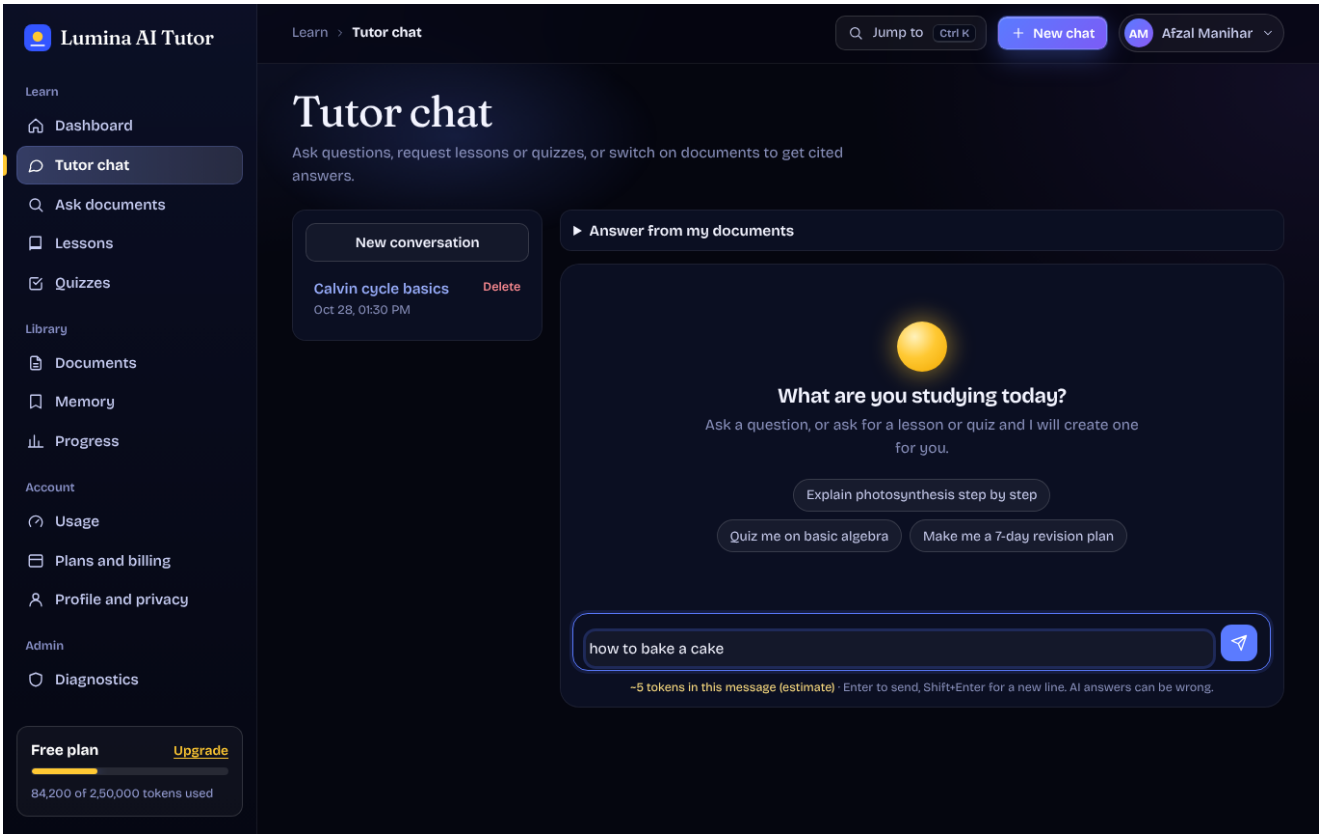
A6 Mobile learning experience



Mobile layouts preserve the same learning actions, readable content, accessible controls, and source awareness.

Product requirement link: use this visual as a design reference for hierarchy and interaction coverage. Final UI must still meet the functional, responsive, accessibility, privacy, and error-state requirements defined in this PRD.

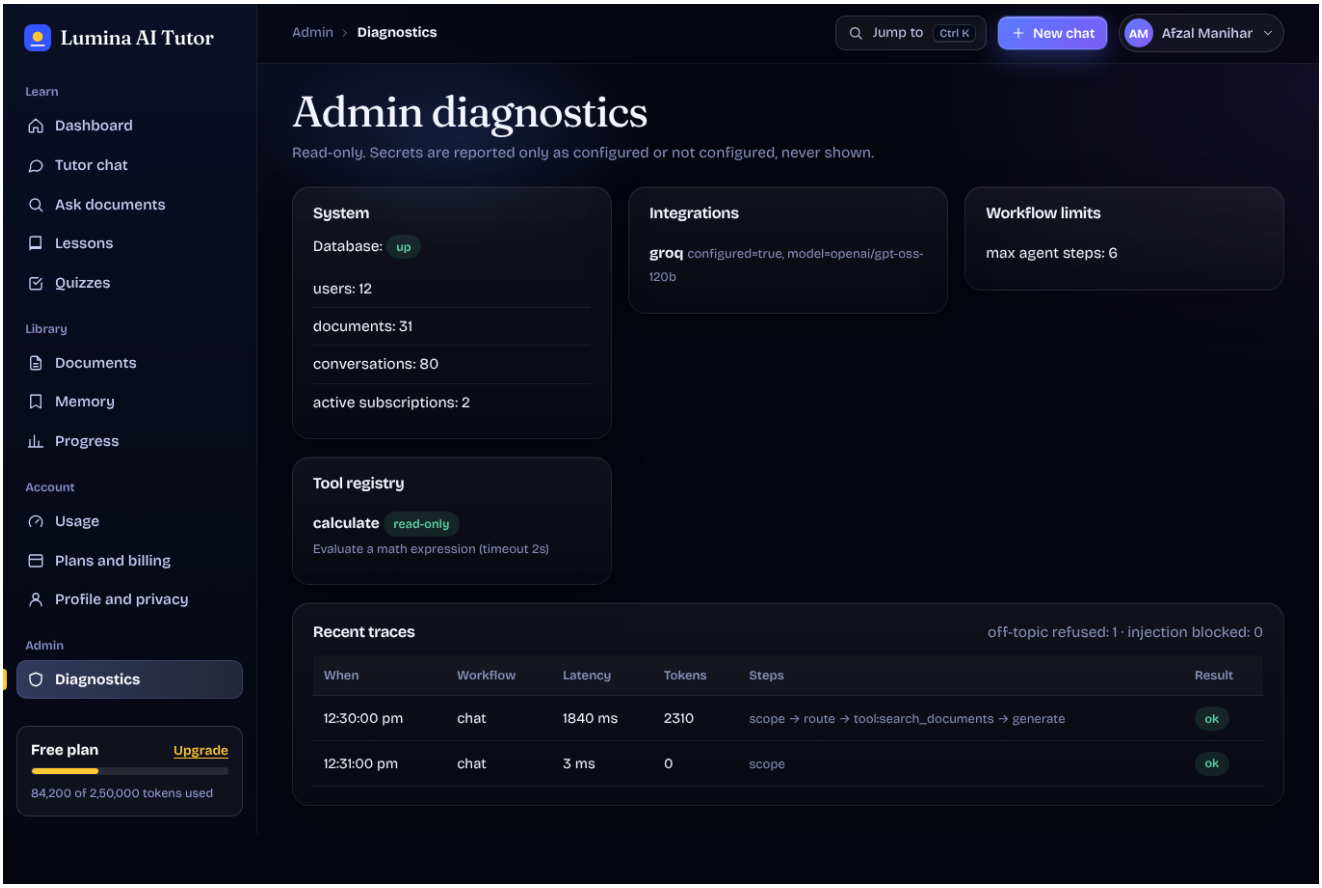
A7 Token and usage transparency



Usage is presented as an estimate or provider-reported value, with period, limit and next action clearly labeled.

Product requirement link: use this visual as a design reference for hierarchy and interaction coverage. Final UI must still meet the functional, responsive, accessibility, privacy, and error-state requirements defined in this PRD.

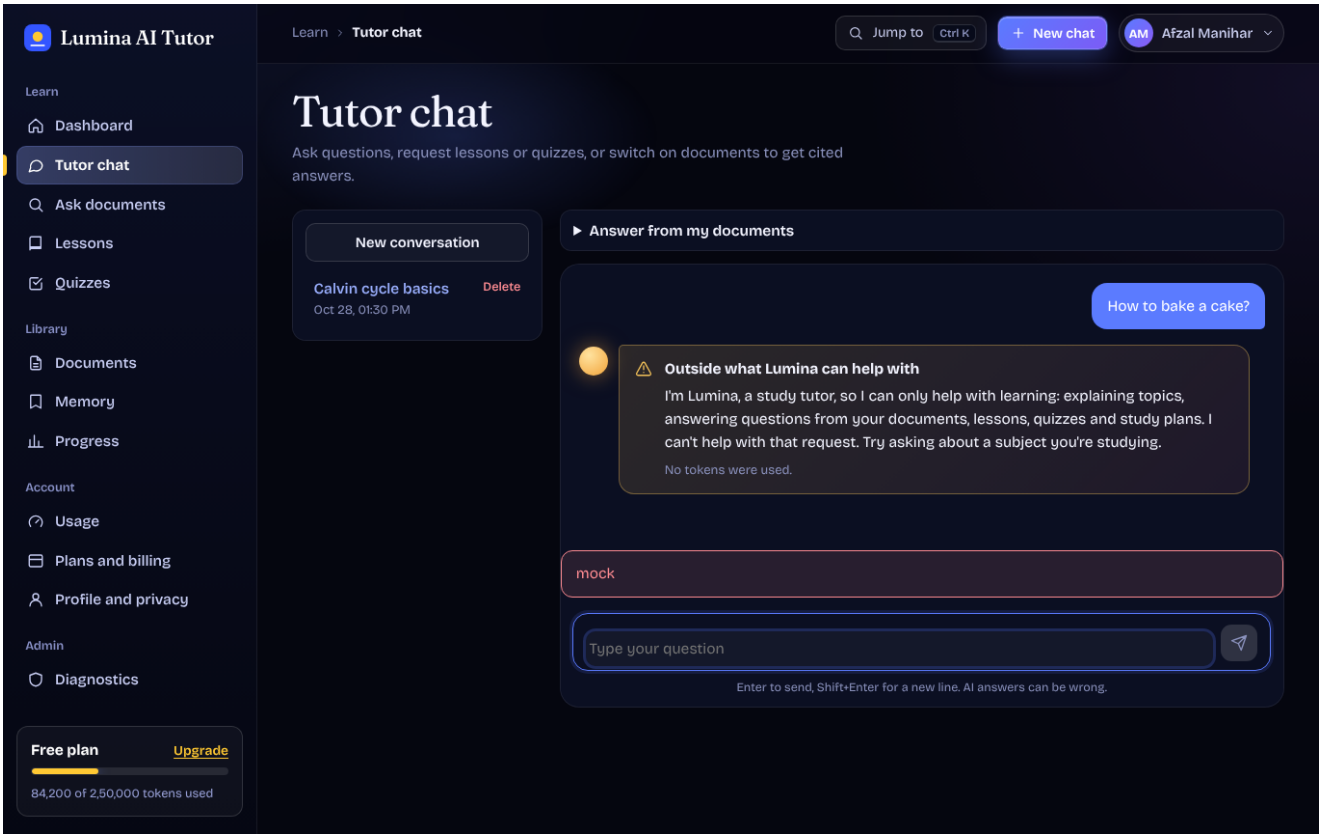
A8 Admin diagnostics and traceability



Operational screens support diagnosis by request ID while keeping secrets and learner content out of default traces.

Product requirement link: use this visual as a design reference for hierarchy and interaction coverage. Final UI must still meet the functional, responsive, accessibility, privacy, and error-state requirements defined in this PRD.

A9 Scope and safety behavior



When a request is outside learning scope, Lumina should set a respectful boundary and offer an in-scope study action.

Product requirement link: use this visual as a design reference for hierarchy and interaction coverage. Final UI must still meet the functional, responsive, accessibility, privacy, and error-state requirements defined in this PRD.

23 Appendix B - Definition of a good first release

A strong first release is not the one with the most screens or the largest number of agents. It is the release in which the learner can reliably complete one private, evidence-grounded learning journey; receive useful adaptive instruction; understand usage and limits; and recover when a provider, file, or network fails.

The release feels complete when...	Proof
The learner trusts document answers.	Citations open the correct page; no-evidence behavior is honest; privacy isolation tests pass.
The learner is taught, not only answered.	Diagnostic, objectives, example, learner attempt, feedback and next step are present.
The AI system is controlled.	Bounded agents/tools, quota, timeout and output validation are observable and tested.
The service can be operated.	A support person can trace an error by ID without private content or credentials.
The product is accessible.	Keyboard/screen-reader/mobile acceptance passes for core journeys.
Commercial behavior is reliable.	Price, payment status and entitlements reconcile on server and provider.

End of Lumina AI Tutor Product Requirements & Production Build Specification - Version 1.0